13B. Construct the DP table for 0/1 knapsack problem and analyze its time and space complexity.

14A. Solve the sum of subsets problem using backtracking and draw the solution tree.

14B. Evaluate the LC Branch & bound method for solving the travelling salesperson problem.

15A Explain nondeterministic algorithms and discuss their role in defining the class NP.

15B. Evaluate the NP completeness of the SAT problem using polynomial reduction.

13B. Illustrate the optimal merge pattern and find the same for ten files whose record lengths are 28, 32, 12, 5, 84, 53, 91, 35, 3, 1

ii) Explain the concept of dynamic programming (DP) in algorithm design provide an example of a problem where DP is applied effectively.

14A. Demonstrate the algorithm for finding m-colouring of a graph & explain with an example.

ii) Explain the N-Queen problem & discuss its algorithm.

14B. Explain the FIFO BB (branch & bound) 0/1 knapsack problem procedure with the knapsack instance for $n=4$, $m=15$, $(P_1, P_2, P_3, P_4) = (10, 10, 12, 18)$ $(w_1, w_2, w_3, w_4) = (2, 4, 6, 9)$

ii) Explain the Branch & Bound technique used in algorithm design. Discuss its key components and how it differs from other algorithmic approaches

15 A. Analyse the concept of NP hardness in the context of algorithm design.

ii Examine the problem of obtaining optimal finish time preemptive schedules for a two processor flow shop is NP hard when jobs are released at two different times R_1 and R_2. jobs released at R_i cannot be scheduled before R_i.

15 B. Compare and contrast NP complete & NP hard problems.

ii) explain Reduction source problem.

13 B. Find the optimal solution to the following 0/1 knapsack problem using dynamic programming technique with knapsack capacity $m=6$, $n=4$

x_i	1	2	3	4	5
p_i	25	20	15	40	50
w_i	3	2	1	4	5

14. A. write an algorithm to solve the graph coloring problem using Backtracking and draw the state space tree with an example graph.

14 B. Find the LC Branch & Bound solution to the travelling salesperson problem with cost matrix as follows

∞	20	30	10	11
15	∞	16	4	2
3	5	∞	2	4
19	6	18	∞	3
16	4	7	16	∞

15.A Discuss NP Hard & NP complete with suitable example.

15B. Differentiate between Deterministic & Non Deterministic algorithms. write a non deterministic search algorithm.

13B. Analyze the Dp method for computing an optimal binary search tree with an example.

14A. solve the graph coloring problem using backtracking & illustrate the color assignment tree.

14B. Evaluate the Branch & Bound approach for solving 0/1 knapsack using cost bound functions.

15A. Explain the characteristics ~~(algorithms)~~ and importance of non deterministic algorithms in complexity theory.

15B. Create a reduction example to prove a problem belongs to NP complete class.

13B. Describe the Dynamic 0/1 knapsack problem. Find an optimal solution for the Dynamic programming 0/1 Knapsack instance for $n=3$, $m=6$ profits are $(P_1, P_2, P_3) = (1, 2, 5)$, weights are $(w_1, w_2, w_3) = (2, 3, 4)$

14A. Develop an algorithm to determine the Hamiltonian cycle in a given graph using backtracking.

14B. Apply branch & Bound algorithm to solve the travelling salesman problem with an example.

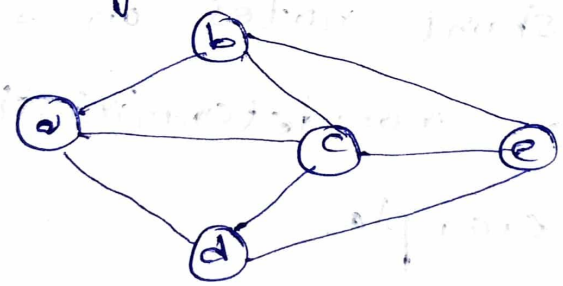
15A. Explain the P, NP, NP Hard and NP complete classes with suitable examples.

15B. Briefly explain cook's theorem.

13B- using algorithm OBST (optimal binary search tree) compute $c(i, j)$ $0 \le i \le j \le 4$ for the identifier set $(a_1, a_2, a_3, a_4) = (do, if, int, while)$ with $(p_1, p_2, p_3, p_4) = (3, 3, 1, 1)$ and $q(0:4) = (2, 3, 1, 1, 1)$ & draw the optimal tree.

14A. Explain the 4-Queens problem using backtracking

ii consider a graph $G = (V, E)$ shown in fig. Find a Hamiltonian circuit using backtracking method.



14B. Draw the portion of the state space tree generated by LCBB for the following knapsack instances.

$n = 5$, $(P_1, P_2, P_3, P_4, P_5) = (10, 15, 6, 8, 4)$

$(w_1, w_2, w_3, w_4, w_5) = (4, 6, 3, 4, 2)$

& $m = 12$

ii) Draw the portion of state space tree generated by FIFO knapsack for the instance

$N = 4$, $(P_1, P_2, P_3, P_4) = (10, 10, 12, 18)$

$(w_1, w_2, w_3, w_4) = (2, 4, 6, 9)$ $m = 15$

15A. Discuss about Cook's theorem in detail

ii) prove the CNF satisfiability of AND/OR graph decision problem.

15B. write short notes on 3-SAT problem

ii) Describe nondeterministic algorithm with an example.

13B. write dynamic programming solution for the travelling sales person problem for the network with the cost adjacency matrix below. Assume node 1 as the home city.

0	10	15	30
4	0	9	11
5	13	0	10
7	7	8	0

14.A. Explain how backtracking is used for solving n-queens problem. Show the statespace tree.

14B. Give the formulation of modified knapsack problem using branch & bound & find the optimal solution using least cost branch & Bound (LCBB) with $n=4$, $m=15$

$(P_1, P_2, P_3, P_4) = (15, 15, 17, 23)$

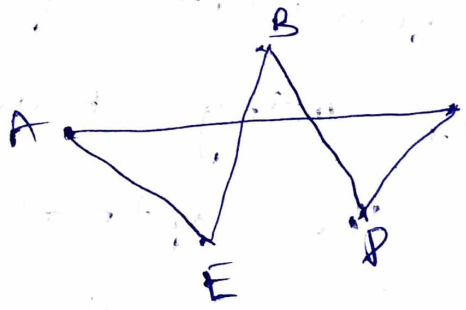
$(w_1, w_2, w_3, w_4) = (3, 5, 6, 9)$

15A. write about nondeterministic algorithms & choice, failure & success functions with search example.

15B. Briefly explain about Np hard & Np complete problems.

13B. Analyse with an example how the branch - bound technique is used to solve 0/1 knapsack problem.

14A. use the branch & bound algorithm to solve the Hamiltonian circuit problem for the following graph



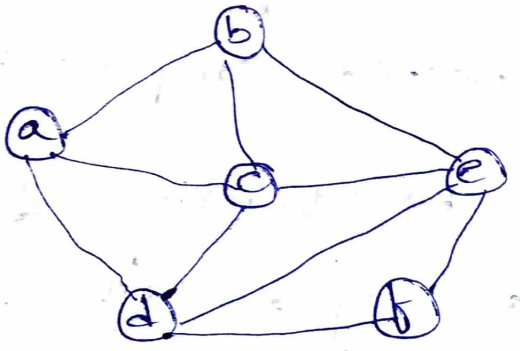
14B. write down & explain the procedure of tackling the 8-queen's problem using a backtracking approach

15A. Examine how non deterministic algorithms relate to NP complete problems.

15B. Describe with an example of a real world problem that is NP hard & explain the challanges associated with solving it.

13B. In the context of optimal binary search trees, analyze how dynamic programming is used to minimize the search cost.

14A. Apply backtracking to the problem of finding a Hamiltonian circuit in the following graph



14B. Discuss the working concept of travelling salesman problem, that could be solved using branch & Bound

technique. Explain with a sample graph.

15A. Summarize the following.

i) Np hard problems

ii) Np complete problems.

15B. Explain the importance of cook's theorem and its contribution to the field of computational complexity.

13B. Consider $n=4$, $M=10$, $P=\{10, 40, 30, 50\}$ $w=\{5, 4, 6, 3\}$ solve it for obtaining optimal solution using 0/1 knapsack.

14A. Explain the following with examples.

i) connected components

ii) biconnected components.

14B. what is backtracking & explain Hamiltonian cycle with example.

15.A what is Branch & Bound and explain with Travelling salesperson problem.

15 B. Explain P & NP class problems

ii) Differentiate NP hard & NP complete problems.